

BOWERS FIELD, WA, USA

WMO: 727883

Lat: 47.034N

Lon: 120.530W

Elev: 1766

StdP: 13.78

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 24220

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.1	12.7	-0.3	5.8	8.7	5.4	7.7	15.3	27.8	41.9	24.3	39.5	1.9	50	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.3	96.8	64.9	92.7	64.0	88.7	62.7	67.4	91.0	65.6	88.5	64.0	85.5	7.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.4	77.7	76.0	56.4	72.5	74.0	54.8	68.1	72.6	32.6	90.3	31.2	88.5	29.9	85.3	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.1	27.7	25.3	1.4	102.1	7.5	3.5	-4.0	104.7	-8.4	106.7	-12.6	108.7	-18.0	111.3		
(5)			0.6	70.5	7.3	2.1	-4.7	72.0	-8.9	73.2	-13.0	74.3	-18.3	75.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	30.3	35.1	42.1	48.7	57.4	64.1	72.1	71.3	61.9	49.4	37.7	29.6		
	DBStd	16.27	8.12	7.14	6.49	6.09	7.30	7.02	6.35	5.91	6.57	6.89	7.90	7.91		
	HDD50	2492	610	418	253	96	18	1	0	0	2	88	372	634		
	HDD65	6028	1075	837	710	490	254	99	14	14	135	483	818	1099		
	CDD50	2512	0	0	7	56	247	424	685	659	359	71	4	0		
	CDD65	576	0	0	0	0	19	73	234	208	41	1	0	0		
	CDH74	7852	0	0	0	38	405	1051	2990	2570	766	32	0	0		
	CDH80	3698	0	0	0	5	135	452	1539	1271	291	5	0	0		
	WSAvg	9.0	4.4	5.6	8.6	11.2	12.2	14.0	13.8	12.2	9.4	7.2	5.3	4.2		
	PrecAvg	9.2	1.3	0.9	0.7	0.7	0.6	0.6	0.3	0.4	0.4	0.6	1.2	1.5		
(15) Precipitation	PrecMax	14.1	4.4	2.2	3.4	2.2	1.7	1.7	1.3	2.1	1.8	1.9	2.6	4.2		
	PrecMin	4.7	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	2.1	0.9	0.6	0.6	0.4	0.4	0.5	0.3	0.5	0.5	0.5	0.7	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.5	57.6	68.2	79.3	89.8	97.6	101.7	101.0	93.3	78.9	61.1	50.2		
		MCWB	44.8	46.7	51.9	55.8	61.5	65.8	65.6	66.3	63.8	58.1	51.4	43.1		
	2%	DB	46.3	52.5	62.4	72.5	83.7	90.6	97.3	96.1	87.8	70.6	56.0	45.8		
		MCWB	40.6	43.8	48.5	53.6	59.8	63.0	64.9	64.6	61.6	54.9	48.5	40.9		
	5%	DB	43.2	49.0	57.3	66.9	78.3	84.8	93.2	91.6	82.6	66.0	52.5	42.8		
		MCWB	38.6	41.4	45.8	51.0	57.5	61.3	64.3	63.8	59.8	52.4	45.7	38.6		
	10%	DB	39.8	45.7	54.0	61.9	72.8	79.9	89.2	87.6	77.9	62.4	49.4	39.4		
		MCWB	36.3	39.6	44.2	48.3	55.4	59.3	63.3	62.6	58.0	50.9	43.6	36.1		
	0.4%	WB	44.7	48.4	53.2	57.9	63.8	68.2	69.5	69.1	66.1	60.1	52.9	44.5		
		MCDB	50.6	56.2	64.9	75.7	85.3	91.1	92.7	92.0	88.2	74.6	58.8	48.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.4	44.5	49.7	54.5	61.2	64.8	67.7	66.9	63.3	56.6	49.6	41.4		
		MCDB	46.0	51.2	60.2	70.8	80.8	87.6	91.3	90.8	83.8	67.5	55.9	45.5		
	5%	WB	38.9	42.2	46.8	51.8	59.0	62.4	65.9	65.2	61.0	54.1	46.6	38.9		
		MCDB	42.7	47.9	56.2	65.3	76.0	82.4	89.0	87.6	79.1	63.8	51.0	42.5		
	10%	WB	36.7	40.1	44.7	49.4	56.7	60.1	64.1	63.6	59.2	51.7	44.3	36.6		
		MCDB	39.5	45.2	52.9	60.5	70.4	77.8	86.0	84.5	74.9	60.8	48.6	39.3		
	5% DB	MDBR	11.4	16.7	21.1	23.7	24.8	25.0	29.3	29.5	28.9	23.7	17.1	11.8		
		MCDBR	16.0	21.7	26.6	32.9	34.9	34.7	37.7	37.9	37.8	30.7	20.4	15.5		
		MCWBR	11.6	14.2	15.6	17.1	15.5	13.9	13.8	14.1	16.2	17.0	13.7	11.6		
		MCWBR	14.8	19.5	24.9	30.6	31.5	32.4	34.9	34.6	34.0	26.5	18.6	14.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.9	13.0	15.2	16.5	14.4	13.5	13.4	13.3	15.4	15.5	13.0	11.4		
	Clear-Sky Solar Irradiance	taub	0.287	0.289	0.298	0.323	0.333	0.333	0.329	0.329	0.320	0.309	0.297	0.283		
		taud	2.462	2.523	2.552	2.462	2.462	2.482	2.508	2.502	2.519	2.541	2.522	2.475		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	258	284	297	296	295	295	295	291	285	272	252	246		
		Edh at Noon	21	24	27	33	34	33	32	31	28	23	20	18		
(44) All-Sky Solar Radiation	RadStd		366	666	1016	1499	1838	2086	2318	1957	1423	839	435	290		
			47	80	92	115	135	136	127	115	96	93	51	42		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+1.30	N/A	N/A	N/A	N/A	N/A	-490	N/A	N/A
(47) Regional (27 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+63

Nomenclature: See separate page